DATA SET DS1530

11. **Rating Political Candidates** A political scientist is studying the relationship between the voter image of a conservative political candidate and the distance (in kilometers) between the residences of the voter and the candidate. Each of 12 voters rated the candidate on a scale of 1–20.

Voter	Rating	Distance
1	12	75
2	7	165
3	5	300
4	19	15
5	17	180
6	12	240
7	9	120
8	18	60
9	3	230
10	8	200
11	15	130
12	4	130

a. Calculate Spearman's rank correlation coefficient r_s.

b. Do these data provide sufficient evidence to indicate a negative rank correlation between rating and distance?

Rank x	Rank y	$d = x - y$
7.5	3	$7.5 - 3 = 4.5$
4	7	$4 - 7 = -3$
3	12	$3 - 12 = 9$
12	1	$12 - 1 = 11$
10	8	$10 - 8 = 2$
7.5	11	$7.5 - 11 = -3.5$
6	4	$6 - 4 = 2$
11	2	$11 - 2 = 9$
1	10	$1 - 10 = -9$
5	9	$5 - 9 = -4$
9	5.5	$9 - 5.5 = 3.5$
2	5.5	$2 - 5.5 = -3.5$

$\Sigma d^2 = 454$

$$r_{s\,STAT} = 1 - \frac{6\Sigma d^2}{n(n^2-1)}$$

$$r_s^* = \frac{6 \cdot 454}{12(12^2-1)} = -0.5874$$

(b). H_0: There is no correlation.

H_a: There is a negative correlation

$n = 12$, $\alpha = 0.05$ $r = 0.497$

$RR: \{ r_s \leq -0.497 \}$.

$-0.5874 < -0.497 \Rightarrow$ Reject H_0

There is sufficient evidence to support the claim of a negative correlation between rating and distance.